

Permutation Matchings in the Order-Preserving Transformation Semigroup

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Abstract

For a positive integer n , let O_n be the semigroup of order-preserving maps from the chain $[n]$ to itself. We prove that there is a bijection $\phi_n : O_n \rightarrow O_n$ for which $\phi_n(f)$ is a mutual generalized inverse of f for every $f \in O_n$. The proof separates the mutual-inverse graph by rank. In a fixed rank, a map is encoded by its image set and the cuts between its consecutive kernel classes. A balanced integer weighting of cut-set/transversal incidences then yields a doubly stochastic weighting supported on the mutual-inverse edges. Hall's theorem supplies a perfect matching in every rank component.

1 Introduction

Write $[n] = \{1, \dots, n\}$, and compose functions from right to left. The order-preserving full transformation semigroup is

$$O_n = \{f : [n] \rightarrow [n] : i \leq j \implies f(i) \leq f(j)\}.$$

Two elements $f, g \in O_n$ are *mutual generalized inverses* if

$$f \circ g \circ f = f, \quad g \circ f \circ g = g. \quad (1)$$

A *permutation matching* of O_n is a bijection of O_n that sends every element to a mutual generalized inverse.

Higgins established a Hall-type characterization of permutation matchings in finite regular semigroups and asked explicitly whether O_n has such a matching [1, Theorem 1.1 and Question 1]. The theorem below answers that question affirmatively for every n . Its central ingredient is not an element-by-element choice rule, but a fractional matching with exact marginals; ordinary finite Hall theory then converts this into a perfect matching.

Our coordinates use uniform subsets and a finite incidence relation. As broad adjacent context for finite set-system incidence questions, one may compare work on uniform intersecting families such as [2]. That work does not concern transformation semigroups, generalized inverses, or the matching proved here, and is not used in the proof.

Theorem 1. *For every integer $n \geq 1$, there exists a bijection $\phi_n : O_n \rightarrow O_n$ such that, for every $f \in O_n$,*

$$f \circ \phi_n(f) \circ f = f, \quad \phi_n(f) \circ f \circ \phi_n(f) = \phi_n(f).$$

2 Rank coordinates and inverse edges

Fix $n \geq 1$ and $1 \leq r \leq n$. Let

$$\mathcal{Q}_{n,r} = \{(y_1 < \cdots < y_r) : y_i \in [n]\}$$

be the set of r -subsets of $[n]$, written in increasing order, and let

$$\mathcal{P}_{n,r} = \{(c_1 < \cdots < c_{r-1}) : c_i \in [n-1]\}.$$

For $C \in \mathcal{P}_{n,r}$, put $c_0 = 0$, $c_r = n$, and define its consecutive blocks by

$$I_i(C) = \{c_{i-1} + 1, \dots, c_i\}, \quad 1 \leq i \leq r.$$

The set $\mathcal{P}_{n,1}$ consists of the single empty tuple.

Every rank- r map in O_n is represented uniquely by a pair $(Y, C) \in \mathcal{Q}_{n,r} \times \mathcal{P}_{n,r}$: if $Y = (y_1 < \cdots < y_r)$, the corresponding map has the constant value y_i on $I_i(C)$. Thus both labelled copies of the rank- r part of O_n have vertex set $\mathcal{Q}_{n,r} \times \mathcal{P}_{n,r}$.

For $C \in \mathcal{P}_{n,r}$ and $X = (x_1 < \cdots < x_r) \in \mathcal{Q}_{n,r}$, write $C \sim X$ when

$$c_{i-1} < x_i \leq c_i \quad (1 \leq i \leq r). \quad (2)$$

Equivalently, X selects exactly one point from each block of C .

Lemma 2 (Inverse criterion). *Let $f = (Y, C)$ and $g = (X, D)$ be rank- r elements of O_n . Then f and g satisfy (1) if and only if*

$$C \sim X \quad \text{and} \quad D \sim Y. \quad (3)$$

Moreover, mutual generalized inverses in O_n necessarily have equal rank.

Proof. If (3) holds, then $f(x_i) = y_i$ and $g(y_i) = x_i$ for every i . For $z \in I_i(C)$, this gives $(f \circ g \circ f)(z) = f(g(y_i)) = f(x_i) = y_i = f(z)$. The other identity follows symmetrically from the blocks of D .

Conversely, (1) implies

$$\text{rank}(f) = \text{rank}(f \circ g \circ f) \leq \text{rank}(g), \quad \text{rank}(g) \leq \text{rank}(f),$$

so the ranks agree. Also $f \circ g$ fixes every point of $\text{im}(f)$, and $g \circ f$ fixes every point of $\text{im}(g)$. Hence the restrictions

$$f|_{\text{im}(g)} : \text{im}(g) \longrightarrow \text{im}(f), \quad g|_{\text{im}(f)} : \text{im}(f) \longrightarrow \text{im}(g)$$

are inverse order-preserving bijections. An order-preserving bijection between two finite chains sends the i -th point to the i -th point. Thus $f(x_i) = y_i$ and $g(y_i) = x_i$, which is precisely (3). $\square$

3 A balanced incidence weighting

The next lemma provides the exact marginals needed for Hall's condition. Multiplicity in its definition is essential.

Lemma 3 (Balanced incidences). *There are nonnegative integers $m(C, X)$, indexed by $C \in \mathcal{P}_{n,r}$ and $X \in \mathcal{Q}_{n,r}$, such that $m(C, X) > 0$ implies $C \sim X$, and*

$$\sum_{C \in \mathcal{P}_{n,r}} m(C, X) = r \quad (X \in \mathcal{Q}_{n,r}), \quad (4)$$

$$\sum_{X \in \mathcal{Q}_{n,r}} m(C, X) = n \quad (C \in \mathcal{P}_{n,r}). \quad (5)$$

Proof. Given $X = (x_1 < \dots < x_r) \in \mathcal{Q}_{n,r}$ and $i \in [r]$, define $C^{(i)}(X) = (c_1^{(i)}, \dots, c_{r-1}^{(i)})$ by

$$c_j^{(i)} = \begin{cases} x_j, & j < i, \\ x_{j+1} - 1, & j \geq i. \end{cases} \quad (6)$$

These cuts lie in $[n-1]$, are strictly increasing, and satisfy $C^{(i)}(X) \sim X$. Define

$$m(C, X) = |\{i \in [r] : C = C^{(i)}(X)\}|. \quad (7)$$

For a fixed X , all r indices in (7) are counted, including repeated cut sets with their multiplicities. This proves (4).

Fix $C \in \mathcal{P}_{n,r}$. For each $i \in [r]$, pairs (X, i) satisfying $C = C^{(i)}(X)$ are in bijection with $t \in I_i(C)$. The subset corresponding to t is uniquely given by

$$x_j = c_j \ (j < i), \quad x_i = t, \quad x_j = c_{j-1} + 1 \ (j > i). \quad (8)$$

It is strictly increasing, and substitution in (6) recovers C . Therefore the number of such pairs is $\sum_{i=1}^r |I_i(C)| = n$, proving (5). This argument also covers $r = 1$, when every cut tuple is empty. $\square$

4 The perfect matching

Proof of Theorem 1. Fix n and r . Form the bipartite graph whose left and right vertex sets are labelled copies of $\mathcal{Q}_{n,r} \times \mathcal{P}_{n,r}$, joining (Y, C) on the left to (X, D) on the right exactly when the two maps are mutual generalized inverses. By Lemma 2, this is equivalent to $C \sim X$ and $D \sim Y$.

Using Lemma 3, assign the nonnegative rational weight

$$w((Y, C), (X, D)) = \frac{m(C, X)m(D, Y)}{nr}. \quad (9)$$

Positive weight is supported on graph edges. For every left vertex (Y, C) , equations (4)–(5) give

$$\sum_{X, D} w((Y, C), (X, D)) = \frac{1}{nr} \left(\sum_X m(C, X) \right) \left(\sum_D m(D, Y) \right) = 1. \quad (10)$$

For every right vertex (X, D) , the same equations in the other order give

$$\sum_{Y, C} w((Y, C), (X, D)) = \frac{1}{nr} \left(\sum_C m(C, X) \right) \left(\sum_Y m(D, Y) \right) = 1. \quad (11)$$

Thus w is a doubly stochastic weighting supported on mutual-inverse edges.

If A is any set of left vertices and $N(A)$ is its graph neighborhood, (10), support on edges, and (11) imply

$$|A| = \sum_{u \in A} \sum_v w(u, v) = \sum_{v \in N(A)} \sum_{u \in A} w(u, v) \leq \sum_{v \in N(A)} \sum_u w(u, v) = |N(A)|.$$

Hall's condition therefore holds. The graph is finite and its two vertex sets have the same size, so it has a perfect matching.

This construction applies separately to every rank $r = 1, \dots, n$. Lemma 2 shows that no mutual-inverse edge joins different ranks. The union of the rank-wise perfect matchings is hence a bijection $\phi_n : O_n \rightarrow O_n$ satisfying the two identities in the theorem. Since $n \geq 1$ was arbitrary, the result follows. $\square$

5 Scope and computational assistance

The argument proves existence through finite Hall theory. It does not select a canonical matching, provide a complexity bound for finding one, make the matching an involution or a semigroup homomorphism, or impose compatibility between different values of n .

During proof development, exact Python checks verified the support and both marginal identities in Lemma 3 for all 78 pairs $1 \leq r \leq n \leq 12$. A separate exhaustive computation constructed the complete mutual-inverse graph of O_6 and a perfect matching. The scripts `interlacing-coupling-check.py` and `n6-inverse-graph.py`, together with their JSON outputs, accompany the paper. These finite checks are corroborative and are not used to infer the unbounded theorem.

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References

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